

Figure 3: Split vector layer dialogue

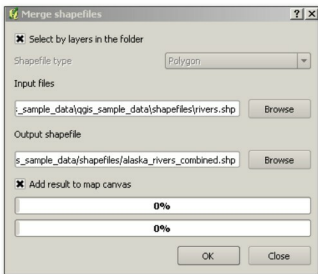


Figure 4: Merge files dialogue

2. In the dialogue box (Figure 2), select the polygon layer to be converted (in our case, add *regions.shp* and select it as the *Input polygon vector layer*).
3. Enter the path and name (e.g., *regions_centroid*) of the new point file.
4. Hit OK and click Yes on the resultant dialogue box to add the output layer to the map canvas.

Split vector layer

This feature allows you to split a vector file into a number of separate shapefiles, based on some attribute value:

1. Click **Vector > Data Management Tools > Split Vector Layer**.
2. Select the *Input vector layer* (see Figure 3) and the attributes to be used to split the layer (in our case, *airports* and the *Unique ID* field, respectively).
3. Browse to the folder to save output files (ours is *airport_split_by_name*) and click OK.

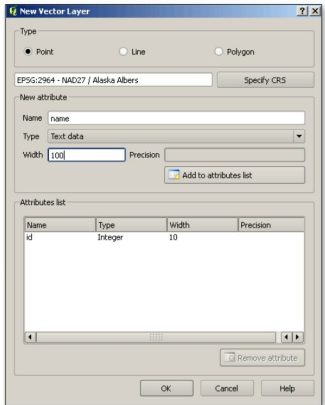


Figure 5: New vector layer dialogue

Merge shapefiles into one

Different shapefiles can be merged into one, provided they are of the same geometry type:

1. Click **Vector > Data Management Tools**.
2. Tick the check-box '*Select by layers in the folder*'. (See Figure 4.)
3. Browse to the source folder, select *majrivers.shp* and *rivers.shp* files.
4. Browse to the target folder and enter the name of the shapefile (*alaska_rivers_combined.shp*).
5. Mark the '*Add result to map canvas*' to directly add the result to the map canvas, and click OK.

Plotting custom features

QGIS allows us to create new layers by plotting geometries and entering attribute data. In a shapefile, one map layer can contain only one type of geometry—point, line or polygon.

Now, let's learn to draw a new point layer by assuming you want to plot sites for new airports. Let this layer have two attributes: *id*, *name*. The *id* attribute is a unique identifier—an integer and added by default.

1. Go to the menu **Layer > New > New Shapefile Layer**.
2. Select **Point** as the geometry (see Figure 5).
3. In the **New attribute** section:
 - i. Add text (e.g., *name*) to the *Name* field.
 - ii. Select **Text data** from the *Type* drop-down.
 - iii. Set *Width* to 100.